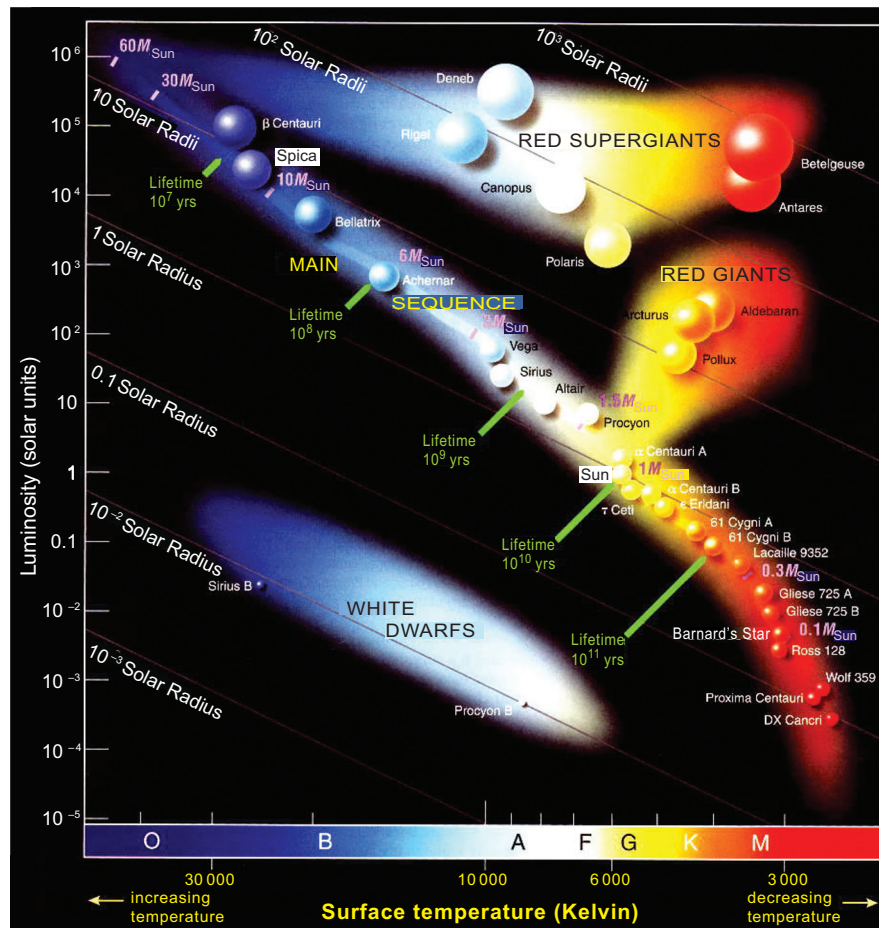


3. One of the most useful diagrams in astrophysics is the Hertzsprung-Russell (HR) diagram. It was developed separately by two astronomers, the Danish astronomer Ejnar Hertzsprung and the American astronomer Henry Norris Russell. Their combined diagram shows that the relationship between the temperature and luminosity of a star is not random and stars fall into distinct groups. These are seen in the Hertzsprung-Russell diagram below which shows the properties of stars.



- (a) Spica is a very large blue star on the main sequence, approximately 10 times the mass of the Sun. Spica is 261 light-years from Earth.

(i) Explain, in terms of forces, why Spica is currently stable. [2]

.....

.....

.....

(ii) Spica is 261 light-years from Earth. State what this means. [1]

.....

.....



Examiner
only

(b) The Sun is also on the main sequence. It is a yellow dwarf star. Explain in terms of forces, the reactions within the Sun and the properties shown on the HR diagram how the Sun will change during the remainder of its life cycle. [6 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

9

